

Task	Prompt	Answer
isomorphic mapping	Given a pair of isomorphic graphs, determine the node correspondence between the two graphs. The first graph is: G describes an undirected graph among 0, 1, 2, 3, 4, 5, and 6. In this graph: Node 0 is connected to nodes 6, 3, 4. Node 1 is connected to nodes 4, 5, 6. Node 2 is connected to nodes 3, 4. Node 3 is connected to nodes 0, 2, 5. Node 4 is connected to nodes 0, 1, 2. Node 5 is connected to nodes 1, 3. Node 6 is connected to nodes 0, 1. The second graph is: G describes an undirected graph among 102, 106, 105, 101, 103, 100, and 104. In this graph: Node 100 is connected to nodes 106, 101. Node 101 is connected to nodes 102, 105, 100. Node 102 is connected to nodes 104, 101, 103. Node 103 is connected to nodes 102, 106, 105. Node 104 is connected to nodes 102, 106. Node 105 is connected to nodes 101, 103. Node 106 is connected to nodes 103, 100, 104. Provide a node matching dictionary such as {Graph1 #Node1: Graph2 #Node1, Graph1 #Node2: Graph2 #Node2, ...}	{0: 102, 3: 101, 2: 105, 4: 103, 1: 106, 5: 100, 6: 104}
jaccard coefficient	The task is to determine the Jaccard coefficient of two nodes in a graph. The Jaccard coefficient is the size of the intersection divided by the size of the union of the neighbors of the two nodes. The input graph is guaranteed to be undirected. Here is an undirected graph containing nodes from 1 to 5. The edges are: (1, 2), (1, 3), (2, 5), (2, 3), (3, 5), (5, 4). Question: What is the Jaccard coefficient between node 2 and node 4? You need to format your answer as a float number.	0.3333
local connectivity	The task is to determine the local connectivity of two nodes in the graph. Local connectivity is whether there exists at least one path between the two nodes. Here is a directed graph containing nodes from 1 to 7. The edges are: (1, 7), (7, 6), (3, 1), (4, 3), (5, 4), (6, 2). Question: What is the local connectivity between node 7 and node 4 in the graph? Your answer should be Yes or No.	No
max weight matching	The task is to determine the maximum weight matching of a graph. A matching is a set of edges without common vertices. A maximal matching cannot add more edges and still be a matching. The weight of a matching is the sum of the weights of its edges. If not specified, all edges have equal edge weights. The input graph is guaranteed to be undirected. Here is an undirected graph containing nodes from 1 to 7. The edges are: (1, 7), (7, 5), (2, 4), (2, 5), (4, 3), (3, 6). Question: What is the maximum weight matching of the graph? You need to format your answer as a list of edges in ascending dictionary order, e.g., [(u1, v1), (u2, v2), ..., (un, vn)].	[(2, 4), (5, 7), (6, 3)]